Frequency Tables

Exam code: 9MA0

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Written by: Amber
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Ungrouped data

- A frequency table for **ungrouped data** shows the frequency of **individual data values**

Why use a frequency table for ungrouped data?

- When collecting large amounts of raw data, it is quicker to use a tally system and then collate the results into a frequency table
- Ungrouped frequency tables are normally used for **numerical, discrete data**
- Organising data into a frequency table makes it easier to work with
  - Calculating averages, ranges and summary statistics can be done much quicker from a frequency table than from raw data
- It gives a clear **pattern** of the data
  - It is easy to quickly see where most of the data are and to see extreme values
- A frequency table for ungrouped data keeps all of the original data values
  - It is still possible to calculate the actual averages, ranges and summary statistics

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Got it

How are the mean, median and mode calculated from an

median and mode from raw, ungrouped data

- The **mode** is the value that occurs **most often** in a data set
  - In an ungrouped frequency table the **data value** with the **highest frequency** will be the mode

- The **median** is the **middle** value when the data is in **order of size**
  - To find the median from an ungrouped frequency table, add the frequencies together until you reach the value that is half of the total
  - For a data set of values, calculate $\frac{n+1}{2}$ and the median will be whichever data value corresponds with this frequency

- The **mean** is the **sum** of all the values **divided by the number of values** in the data set
  - To find the mean, multiply each data value by its corresponding frequency, find the sum of these values and then divide by the total frequency
  - The notation for the sum of the values is Σxf
  - The formula for the mean from a frequency table is $\frac{\Sigma xf}{\Sigma f}$

Help me understand

Worked Example

The frequency table below gives information about the shoe size of a group of year 12 students.

Shoe Size	37	37.5	38	38.5	39	39.5	40	40.5	4
Frequency	3	3	12	5	16	9	7	4	1

(i) Write down the modal shoe size.

(ii) Explain how you know that the median shoe size is 39.

(iii) Calculate the mean shoe size.

Answer:

(ii) To find the median we first need to know the total number of students:

$$f = 3 + 3 + 12 + 5 + 16 + 9 + 7 + 4 + 1 = 60$$

The data is discrete so we use $\frac{n+1}{2}$ to find which value the media will be.

$$\frac{60+1}{2} = 30.5^{th} \text{ value}$$

Use the cumulative frequencies to find which data value is half way between the 30th and the 31st.

Shoe size	37	37.5	38	38.5	39	39.5	40	40.5	41
Frequency	3	3	12	5	16	9	7	4	1
Cumulative frequency	3	6	18	23	39				

+3 +12 +5 +16

We are looking for the 30.5^{th} value so we don't need to go further

The median shoe size is half way between the 30th and 31st shoe size which are both 39, therefore median = 39.

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(iii)

Shoe size, x	37	37.5	38	38.5	39	39.5	40	40.5	41
Frequency, f	3	3	12	5	16	9	7	4	1
xf	111	112.5	456	192.5	624	355.5	280	162	41

$$\text{mean} = \bar{x} = \frac{\Sigma xf}{\Sigma f} = \frac{111 + 112.5 + 456 + \ldots + 162 + 41}{60} = \frac{2334.5}{60}$$

Mean = 38.9 (3.s.f.)

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Examiner Tips and Tricks

- Use common sense when checking your answers, is your mean within the range of the data? Does is seem right? A mean of 140 for example could not be correct if the data is about ages of students taking an exam at university.

Did this video help you? Yes No

Grouped data

- A frequency table for **grouped data** is usually used for large amounts of **continuous data**

What are the advantages and disadvantages of grouping data?

- Grouping data is especially useful when data can take a large range of different values
- Trends and patterns can be easily spotted when data has been grouped
- Calculations are much quicker with data that has been grouped
- It is important to be aware, however, that grouped frequency tables also have some negatives
  - The actual data values are lost when data is grouped
  - It is only possible to calculate estimated averages, ranges and summary statistics

What notation is used for grouped frequency tables?

- When grouping data, it is important to be clear about which **group** or **class** any data value should be entered into
  - A group entry of 10 – 20 followed by 20 – 30 would be unclear because the number 20 could be entered into both groups
    - If the data are discrete, this could be written as 10 – 19 and 20 – 29
    - For continuous data, this could be changed to 10 – and 20 –
    - This would most likely be represented as $10 \leq x < 20$ and $20 \leq x < 30$

- Most commonly inequalities are used to group continuous data as they leave no ambiguity
- If the data are continuous, always check that there are no gaps between upper boundary of a class and the lower boundary of the next class
  - If there are gaps you will need to close these gaps by changing the boundaries before carrying out any calculations
    - For example, the groups $10 \leq x \leq 19$ followed by $20 \leq x \leq 29$ will become $9.5 \leq x < 19.5$ and $19.5 \leq x < 29.5$
    - Check the inequality signs carefully

- Be careful when deciding what category the data falls into, taking the group $10 \leq x \leq 19$ for example,
  - if the data had been rounded it would take the form $9.5 \leq x < 19.5$ as described above
  - if the data had been **truncated** however, then the boundaries would become $10 \leq x < 20$
    - this is most likely to happen with age, which is technically a continuous variable due to being able to take any value, however we would usually consider age by counting years

How do I find averages from grouped frequency tables?

- Instead of finding the **mode**, when working with a grouped frequency table we instead find the **modal class**

and the **median** from a grouped frequency table
- Calculating the **estimated mean** is the same as for ungrouped frequency tables, however you will need to find the **midpoints** first
  - the midpoint is the mean of the upper and lower values in the class boundaries
  - multiply the midpoint by its corresponding frequency, find the sum of these values and then divide by n

- Calculating the **estimated median** is more complicated and can involve using **linear interpolation**

How do I use linear interpolation on grouped data?

- For grouped data, the **median, quartiles and percentiles** can be found using a process called **linear interpolation**
- **Interpolation** uses the assumption that the data values are evenly spread throughout each class
  - Once you know which data value you are looking for, interpolation uses proportion to find how far through each class the data value should be
  - It is important to remember to add on the lower-class boundary after finding the correct data value

- Follow these steps to use linear interpolation to find the median, quartiles or percentiles:

Step 1. Find the place within the data where the value you are looking to find is positioned, e.g. the median will be the $\frac{n}{2}$ th value

Step 2. Add up the frequencies until you know which class this value will fall into (this is finding the cumulative frequency)

Step 3. Write the upper- and lower-class boundaries of this group on the upper side of a simple number line

Step 4. On the other side of the same number line add the lower and upper values of the **cumulative frequency** for this group to the ends of the line

Step 5. Use proportion to calculate the data value that is the same fraction of the way through the group on the upper side of the number line as where the value you are looking for would sit on the lower side of the number line

- To see these steps in action, go to part (iv) of the worked example
- This method can be used to find any percentile, for example the 81st percentile will be the value that is $\frac{81}{100}$ ths of the way through the data

Worked Example

The table below shows the heights in cm of a group of year 12 students.

$155 \leq h < 160$	5
$160 \leq h < 165$	9
$165 \leq h < 170$	7
$170 \leq h < 175$	1

(i) Write down the class that a student of exactly 160 cm should be added to.

(ii) Write down the modal class.

(iii) Calculate the estimated mean height.

(iv) Use linear interpolation to find the interquartile range of the heights

Answer:

Examiner Tips and Tricks

- There can be a lot of calculations when working with grouped frequency tables, be extra careful when using your calculators as it is easy to make small errors with these questions. Use the table and add information to the table as you go as there will be marks available for showing work within the methods. Make sure you know how to use your calculator to find all of the summary statistics as this can save time and reduce errors.

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